Weak-star PCP is basis-determined over weak-star angelic preduals

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Status and source problem

This packet records a likely-valid substantial partial result. The source is G. López-Pérez and J. A. Soler-Arias, *Weak-star point of continuity property and Schauder bases*, arXiv:1309.3862 [1]. In its introduction the paper recalls Bourgain's open problem asking whether PCP, respectively RNP, is determined by subspaces with a Schauder basis. Its Corollary 2.10 proves the weak-star analogue for a norm-closed subspace of the dual of a *separable* Banach space. The same paper explicitly notes that it does not know whether weak-star PCP is separably determined in general.

> We recall that a closed and bounded subset of a Banach space X satisfies the point of continuity property (PCP) if every closed subset of C has some point of weak continuity, that is, the weak and the norm topologies agree at this point. Also, when X is a dual space, C is said to satisfy the weak-star point of continuity property (w^*-PCP) if every closed subset of C has some point of weak-$*$ continuity, equivalently every nonempty subset of C has relatively w^*-open subsets with diameter arbitrarily small. X has PCP (resp. w^*-PCP when X is a dual space) if B_X, the closed unit ball of X, has PCP (resp. w^*-PCP). Also, a subspace X of a dual space Y^* is said to verify the w^*-PCP if B_X, as a subset of Y^*, has the w^*-PCP. It is well known that RNP implies PCP, being false the converse, and it is clear that w^*-PCP implies PCP. Moreover, RNP and w^*-PCP are equivalent for convex w^*-compact sets in a dual space, see theorem 4.2.13 in [2]. We will use this last fact freely in the future. We refer to [9] for background about PCP and w^*-PCP. It is a well known open problem [1] if PCP (resp. RNP) is determined by subspaces with a Schauder basis. Our goal is characterize w^*-PCP for closed and bounded subsets of dual spaces of separable Banach spaces and conclude in theorem 2.10 that, in fact, w^*-PCP is determined by subspaces with a Schauder basis in the natural setting of subspaces of dual spaces with a separable predual. As an easy consequence we also deduce from the above characterization of w^*-PCP that every seminormalized basic sequence in a Banach space with PCP has a boundedly complete basic subsequence. This last result was obtained in [8].
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> We begin with some notation and preliminaries. Let X be a Banach space and let B_X, respectively S_X, be the closed unit ball, respectively

The result below replaces separability of the predual by the broader assumption that its dual ball is weak-star Fréchet–Urysohn. Such Banach spaces are commonly said to have weak-star angelic dual. In particular, all weakly Lindelöf determined (WLD) spaces have weak-star angelic dual [2].

Definitions

Let Y be a Banach space. All weak-star topologies on subspaces of Y^* are the relative topology inherited from $\sigma(Y^*, Y)$. A norm-closed bounded set $K \subseteq Y^*$ has the *weak-star point of continuity property* (w^*-PCP) if every nonempty norm-closed subset of K has a point at which the identity from the weak-star topology to the norm topology is continuous. Equivalently, every nonempty subset of K has relatively weak-star open subsets of arbitrarily small norm diameter.

A topological space is *Fréchet–Urysohn* if, whenever $t \in \overline{A}$, there is a sequence in A converging to t. We say that Y has *weak-star angelic dual* when (B_{Y^*}, w^*) is Fréchet–Urysohn.

A tree $(x_A)_{A\in\mathbb{N}^{<\omega}}$ is *weak-star null* if $x_{(A,n)} \to 0$ weak-star for every A. It is *seminormalized* if

$$0 < \inf_A \|x_A\| \le \sup_A \|x_A\| < \infty.$$

Its partial sum at A is $p_A = \sum_{C\le A} x_C$. A full subtree is obtained by passing, at every node, to a subsequence of its immediate successors.

Idea of the proof

Failure of w^*-PCP supplies a bounded set B with no relatively weak-star open subset of small diameter. Thus every $b \in B$ lies in the weak-star closure of points of B a fixed norm distance away. The Fréchet–Urysohn hypothesis turns each such closure statement into a convergent sequence. Iterating and taking parent–child differences gives a seminormalized weak-star-null tree whose partial sums telescope inside B.

The remaining issue is to put all nodes into one basis subspace. Enumerate the tree compatibly with its order. At each stage choose the next successor arbitrarily close to a nested weak-star closed finite-codimensional subspace which annihilates a finite norming set for the span of the preceding vectors. After a summable perturbation, the chosen vectors have uniformly bounded partial-sum projections and therefore form a basic sequence. Fullness keeps the original branchwise partial sums, so the generated Schauder-basis subspace still fails w^*-PCP.

Tree characterization

Proposition 1. *Let Y have weak-star angelic dual and let $K \subseteq Y^*$ be norm-closed and bounded. The following are equivalent.*

1. *K fails w^*-PCP.*
2. *There is a seminormalized weak-star-null tree $(x_A)_{A\in\mathbb{N}^{<\omega}}$ such that*

$$\left\{ \sum_{C\le A} x_C : A \in \mathbb{N}^{<\omega} \right\} \subseteq K.$$

Proof. Assume first that K fails w^*-PCP. By the standard fragmentation formulation used in [1, Proposition 2], there are $B \subseteq K$ and $\delta > 0$ such that every nonempty relatively weak-star open subset of B has diameter greater than 2δ. Hence, for every $b \in B$,

$$b \in \overline{B \setminus B(b,\delta)}^{w^*}.$$

Choose R with $B \subseteq RB_{Y^*}$. Scalar multiplication identifies RB_{Y^*} with B_{Y^*} topologically, so it is Fréchet–Urysohn. Consequently, for every $b \in B$ there is a sequence $(b_n)_n \subseteq B \setminus B(b, \delta)$ converging weak-star to b.

Since $\operatorname{diam} B > 2\delta$, choose the initial point $y_\varnothing \in B$ with $\|y_\varnothing\| > \delta$. Iterating the sequential choice gives a tree $(y_A)_{A \in \mathbb{N}^{<\omega}}$ in B such that $y_{(A,n)} \to y_A$ weak-star and $\|y_{(A,n)} - y_A\| > \delta$. Define

$$x_\varnothing = y_\varnothing, \qquad x_{(A,n)} = y_{(A,n)} - y_A.$$

The tree is weak-star null at every node and seminormalized: the root and all non-root terms have norm greater than δ, and all terms are bounded because B is bounded. Finally the sum telescopes:

$$\sum_{C \le A} x_C = y_A \in B \subseteq K.$$

Conversely, suppose (2) holds and put $d = \inf_A \|x_A\| > 0$. Write $p_A = \sum_{C \le A} x_C$. For each A,

$$p_{(A,n)} = p_A + x_{(A,n)} \longrightarrow p_A \quad (w^*), \qquad \|p_{(A,n)} - p_A\| \ge d.$$

It follows that every nonempty relatively weak-star open subset of $P = \{p_A : A \in \mathbb{N}^{<\omega}\}$ has diameter at least d. The norm closure $\overline{P}^{\|\cdot\|}$ is a norm-closed subset of K. Since weak-star open sets are norm-open and P is norm-dense in this closure, the same diameter lower bound holds there. Thus K fails w^*-PCP. □

Full basic subtree lemma

Lemma 2. *Let Y be any Banach space and let $(x_A)_{A \in \mathbb{N}^{<\omega}} \subseteq Y^*$ be a seminormalized weak-star-null tree. It has a full weak-star-null subtree whose nodes, in an enumeration which places parents before children, form a Schauder basic sequence.*

Proof. Fix an order-compatible enumeration $(A_n)_{n \ge 1}$ of $\mathbb{N}^{<\omega}$. Put $c = \inf_A \|x_A\| > 0$. We construct a full subtree with enumerated nodes (u_n), nearby vectors (v_n), and decreasing weak-star closed finite-codimensional subspaces

$$Y^* = M_0 \supseteq M_1 \supseteq M_2 \supseteq \cdots .$$

Choose $0 < \eta < 1/2$ and set $K_0 = (1-\eta)^{-1}$. Choose positive $(\varepsilon_n)_{n \ge 2}$ so that

$$\sum_{n \ge 2} \varepsilon_n < \frac{c}{8K_0} \quad \text{and} \quad \varepsilon_n < c/2.$$

Set $u_1 = v_1 = x_\varnothing$. Suppose $u_1, \dots, u_{n-1}$, $v_1, \dots, v_{n-1}$, and M_{n-1} have been chosen. The label A_n is an immediate successor of a label already treated. In the corresponding successor sequence of the original selected parent, take an unused successor u_n far enough out that

$$\operatorname{dist}(u_n, M_{n-1}) < \varepsilon_n.$$

This is possible because the successor sequence is weak-star null and the quotient Y^*/M_{n-1} is finite-dimensional: the quotient map sends that sequence to a norm-null sequence. Choose $v_n \in M_{n-1}$ with $\|u_n - v_n\| < \varepsilon_n$. Requiring increasing original successor indices at each node makes (u_A) a full subtree; its successor sequences remain weak-star null.

Let $E_n = [v_1, \dots, v_n]$. By compactness of the unit sphere of E_n, there is a finite set $F_n \subseteq B_Y$ which $(1-\eta)$-norms E_n:

$$\max_{y \in F_n} |v(y)| \geq (1-\eta)\|v\| \qquad (v \in E_n).$$

Define

$$M_n = M_{n-1} \cap \bigcap_{y \in F_n} \ker y.$$

For $v \in E_n$ and $m \in M_n$ this gives

$$\|v\| \leq K_0 \|v + m\|. \tag{1}$$

Every v_j with $j > n$ belongs to M_n. Hence (1), applied to arbitrary finite linear combinations, proves that (v_n) is a basic sequence with basis constant at most K_0.

Moreover $\|v_n\| \geq c/2$. If (v_n^*) are its coordinate functionals, the difference of two partial-sum projections gives

$$\|v_n^*\| \leq \frac{2K_0}{\|v_n\|} \leq \frac{4K_0}{c}.$$

Therefore

$$\sum_n \|u_n - v_n\| \, \|v_n^*\| < \frac{1}{2}.$$

The small perturbation principle now shows that (u_n) is equivalent to (v_n) and is itself a Schauder basic sequence. This is the desired full basic weak-star-null subtree. □

Basis determination

Theorem 3. *Let Y have weak-star angelic dual, and let X be a norm-closed subspace of Y^*. Then the following are equivalent.*

1. *X has w^*-PCP.*
2. *Every norm-closed subspace of X having a Schauder basis has w^*-PCP (for the topology inherited from Y^*).*

Proof. The forward implication is hereditary: a norm-closed subset of the unit ball of a closed subspace is also a norm-closed subset of B_X.

For the converse, suppose X fails w^*-PCP. Apply Proposition 1 to $K = B_X$. We obtain a seminormalized weak-star-null tree (x_A) whose partial sums lie in B_X. By Lemma 2, pass to a full basic subtree (u_A). Let

$$Z = \overline{\operatorname{span}}\{u_A : A \in \mathbb{N}^{<\omega}\}.$$

The order-compatible enumeration of the nodes is a Schauder basis for Z.

Fullness preserves ancestry. Thus, if θ is the embedding of the new tree into the old one, then

$$q_A := \sum_{C \leq A} u_C = \sum_{D \leq \theta(A)} x_D \in B_X \cap Z.$$

For every A, $q_{(A,n)} \to q_A$ weak-star while $\|q_{(A,n)} - q_A\| = \|u_{(A,n)}\|$ is bounded below uniformly. Exactly the converse argument in Proposition 1 shows that the norm closure of $\{q_A\}$ is a closed subset of B_Z with no weak-star-to-norm point of continuity. Hence Z fails w^*-PCP. This contradicts (2). □

Corollary 4. *Theorem 3 holds whenever Y is WLD. In particular it holds whenever Y is weakly compactly generated, with no separability assumption.*

Proof. WLD Banach spaces have weak-star angelic dual; equivalently, their weak-star dual balls are Corson compact and therefore Fréchet–Urysohn [2]. Weakly compactly generated spaces are WLD. □

Verification, limitations, and novelty status

The proof is qualitative and uses no numerical experiment. The two points most in need of expert checking are: (i) that weak-star closed finite-codimensional quotients carry their usual finite-dimensional topology, which justifies the distance selection in Lemma 2; and (ii) that the full-subtree embedding preserves the complete ancestor chain, which justifies the partial-sum identity in Theorem 3. Both are made explicit in the proof.

This does *not* solve the unrestricted intrinsic PCP/RNP problem. It strengthens the source's weak-star theorem from separable preduals to weak-star angelic preduals, including all WLD preduals. The proof does not extend as written to a predual whose dual ball is merely weak-star sequential: a specified closure point need not be the limit of a sequence from the set, so the null-tree construction may require transfinite sequential closure.

Bounded novelty checks searched the run indexes for arXiv:1309.3862 and the terms 'topologically weak-star null', 'weak-star PCP', 'WLD', 'weak-star angelic', and 'determined by subspaces with a Schauder basis'. Exact-phrase and close-variant arXiv/web searches found [1], its tree precursor, and later work on weak-star sequential dual balls such as [2], but no explicit WLD/weak-star-angelic extension of the basis-determination theorem. The novelty confidence is moderate; the packet should be reviewed as a new partial theorem rather than as a literature-status identification.

References

[1] G. López-Pérez and J. A. Soler-Arias, *Weak-star point of continuity property and Schauder bases*, Studia Math. 219 (2013), 225–236; arXiv:1309.3862.

[2] G. Martínez-Cervantes, *Banach spaces with weak-star sequential dual ball*, Proc. Amer. Math. Soc. 146 (2018), 1829–1835; arXiv:1612.05948.

[3] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I: Sequence Spaces*, Springer, 1977.